

Farmer Friend (Hackathon Project Documentation)

Team: Modern Minds

Team Members:

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1. Problem Statement

In India, many farmers now have access to smartphones and internet connectivity, but they still struggle to use modern agricultural tools because most applications are difficult to understand and not designed for non-technical users.

Farmers often face:

- Crop losses due to lack of disease awareness
- Difficulty understanding complex farming tools
- Limited access to affordable expert guidance
- Problems understanding English-based applications
- Lack of easy access to local weather information

Most existing platforms either feel too technical or have outdated user interfaces that make farmers avoid using them.

Our goal was to build a simple and userfriendly farming assistant that makes modern technology accessible to farmers without confusion.

2. Solution Approach

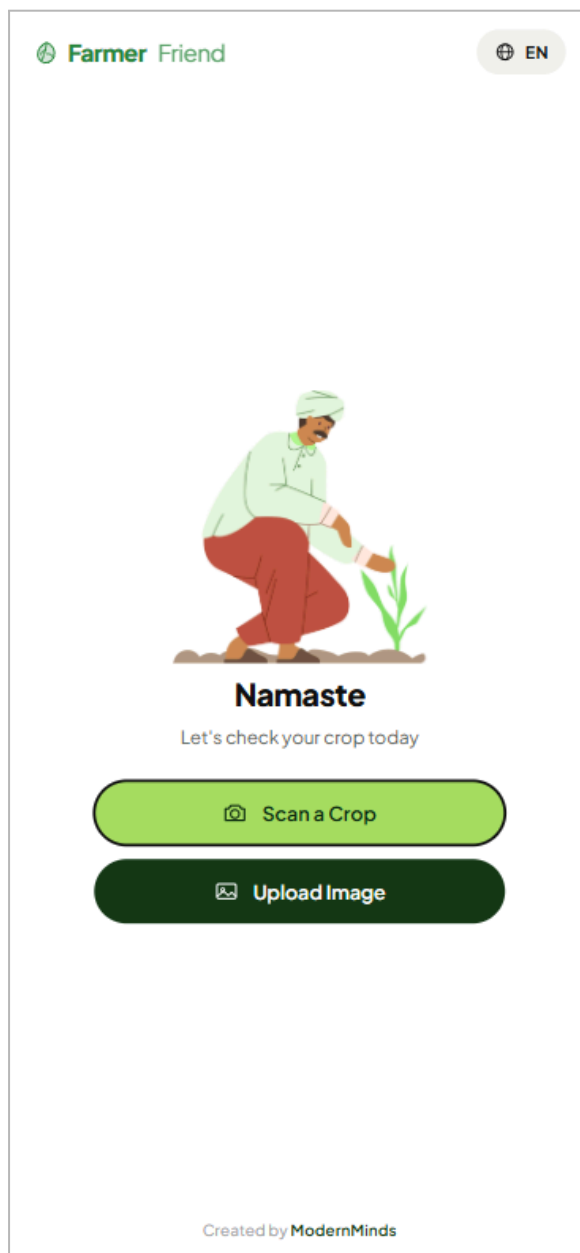
We developed Farmer Friend, a web-based mobile friendly farming assistant specially designed for Indian farmers.

The application focuses on simplicity and accessibility.

Main approach used in the project:

1. **Simplified UI Design** We designed a clean and modern interface with large buttons, simple navigation, and minimal complexity.
2. **AI-Based Crop Assistance** Farmers can upload crop images and receive simple explanations about diseases and treatments.
3. **Voice Assistance** We implemented a speaking assistant to help farmers who may not be comfortable reading text.
4. **Weather Forecast Integration** The app provides regional weather forecasts so farmers can make better farming decisions.
5. **Free Accessibility** We used free API integrations and optimized the application so the platform remains free to use.

Home Page UI Design:



3. System Architecture

Frontend

- HTML
- CSS
- JavaScript

Design Tools

- Figma
- Claude Code

APIs Used

- Google Gemini

Development Support

- Claude Code (API integration, debugging, optimization)
- Senior Mentor: Helped in advance development.

Workflow

User Input → App Processing → API Response → Simplified Output to Farmer

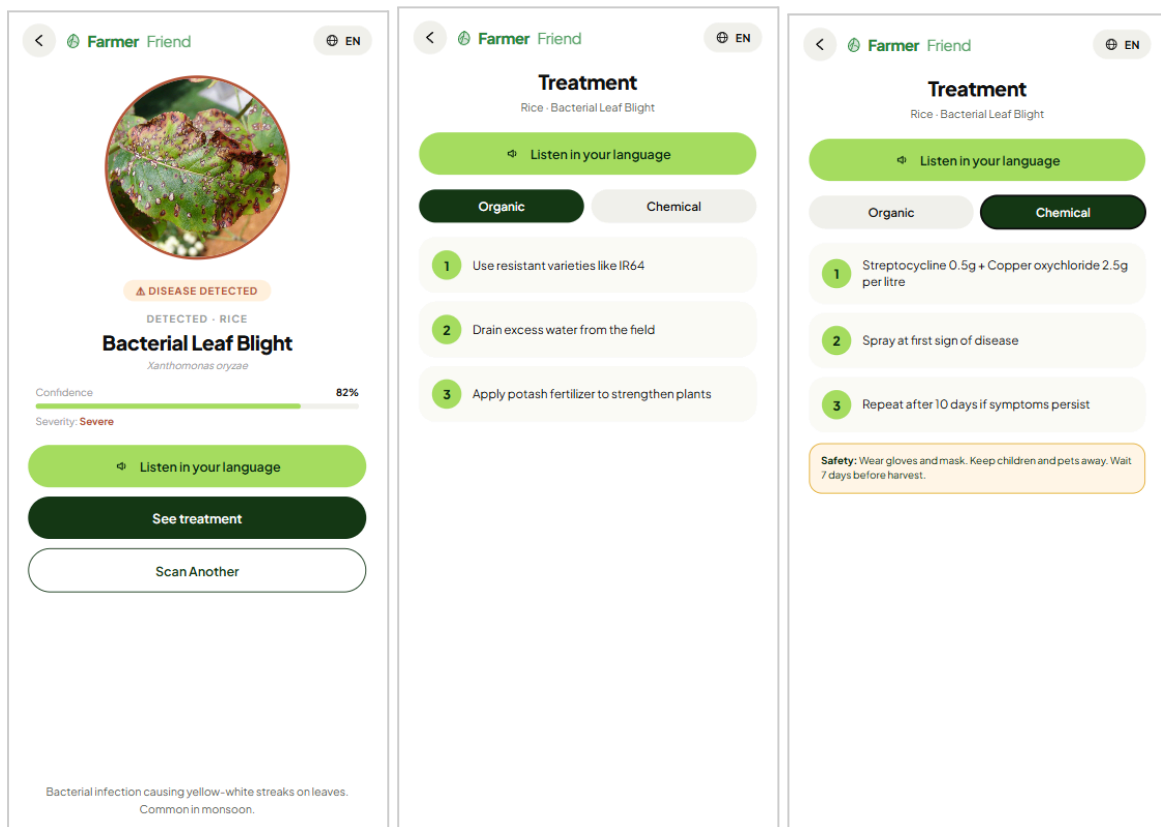
Live Demo👉: <https://f.io/U6z4jSwK>

4. Features

4.1 Crop Disease Detection

Users can upload crop images and receive:

- Disease identification
- Simple treatment suggestions
- Prevention tips



4.2 Voice Assistance

The application includes a voice based assistant to help farmers interact with the app more easily.

(Currently not functioning)

4.3 Simple User Interface

The app is designed with simplicity as the main focus.

Features:

- Large buttons
- Minimal text
- Mobile-friendly layout
- Easy navigation

5. Challenges Faced

5.1 API Integration

Connecting multiple APIs together and managing responses properly was challenging.

5.2 Simplifying the User Experience

One of the biggest challenges was reducing complexity while still keeping useful features.

5.3 Responsive Design

Making the application work properly on different mobile screen sizes required multiple frontend adjustments.

5.4 Understanding Real User Needs

We had to think from a farmer's perspective instead of a technical perspective.

6. Team Contributions

Member 1

- Project ideation
- UI/UX design in Figma
- Presentation planning
- User experience improvements

Member 2

- Frontend styling
- CSS implementation
- Responsive design improvements
- Layout optimization

Member 3

- Backend logic
- API integration

We all three contributed in the entire app development as we don't have advance coding knowledge.

8. Conclusion

FarmerFriend was created with the vision of making modern agricultural technology simple and accessible for Indian farmers.

Instead of focusing only on advanced technology, we focused on usability, simplicity, and real-world impact.

We believe technology becomes truly useful only when everyone can use it easily.

Thank You.

